

IT2140- Database Design and Development

Year 2, Semester 1

Lab Sheet 01: SQL Data Retrieval and Filtering using SELECT, WHERE, ORDER BY, GROUP BY, and HAVING Clauses (Single-Table Queries)

Objective

At the end of this lab session, you should be able to write the **SELECT** command with the **WHERE**, **ORDER BY**, aggregate functions, **GROUP BY**, and **HAVING** clauses for single-table queries.

Exercises (Section 1) — see Data_Set.sql for reference

Write SQL queries for the following:

- a. Retrieve all information of all employees.
- b. Display the names and salaries of employees who earn more than 50,000.
- c. Find employees whose names start with the letter 'A'.
- d. List the names and ages of employees younger than 30.
- e. Find the employees whose salary is between 20,000 and 60,000.
- f. Show employee names whose names contain the letter 'a'.
- g. Display the employee ID and name of those who are either aged 24 or 28.
- h. Select employees whose age is **not** between 25 and 35.
- i. List all employees whose salary is not null.

Exercises (Section 2) — see Data_Set.sql for reference

Write SQL queries for the following:

- j. List all employees sorted by their name (ename).
- a. Show all employees sorted in descending order of salary.
- b. Display employee details ordered ascending by age and descending by salary.
- c. Show all departments sorted by budget in descending order.
- d. List all work assignments ordered by pct_time.

Exercises (Section 3) — see Data_Set.sql for reference

Write SQL queries for the following:

- a. How many employees are in the organization?
- b. What is the total salary paid to all employees?
- c. What is the highest salary and the lowest salary among all employees?
- d. What is the average salary of employees?
- e. How many employees are aged below 30?
- f. What is the total number of departments?
- g. What is the maximum and minimum department budget?
- h. What is the total percentage of work assigned across all employees? (works table)
- i. What is the average work time (pct_time) per employee?

Exercises (Section 4) — see Data_Set.sql for reference

Write SQL queries for the following:

- a. What is the total salary paid to employees of each age group?
- b. Count the number of employees working in each department. Rename the count as 'Number of Employees'.
- c. How many employees earn a salary greater than 50,000?
- d. List all employees who earn more than 50,000, sorted by salary in ascending order.

Exercises (Section 5) — see Data_Set.sql for reference

Write SQL queries for the following:

- a. Display the department IDs where the number of employees is less than 3.
- b. List each manager and the number of departments they manage. Show only managers who manage more than 1 department. Sort by the number of departments ascending.
- c. Show each age group and the average salary of employees in that group. Display only age groups where the average salary is greater than 40,000.
- d. Find the number of work assignments per employee. Display employees with more than 2 assignments.
- e. List departments with a total budget greater than 1,000,000.

Submission Requirement:

Complete **Section 5** of this lab sheet and upload your SQL query answers to the **Git repository** provided by your instructor **before end of the day**.

File Naming Guidelines:

- Name your submission file as:

Lab_1_Exercises_Section_5.sql

Additional Instructions:

- Ensure your SQL queries are well-formatted and include comments where necessary.
- Test your queries before uploading to confirm they run without errors.
- Commit and push your file to the correct Git repository and branch as instructed.